

Charles Neau

Senior / Lead Software Engineer & Systems Architect

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Professional Summary

Versatile, hands-on Lead Software Engineer and Systems Architect with 10+ years of experience engineering high-performance distributed platforms, discrete-event simulation & logistics optimization engines, and full-stack cloud-native web applications. Proven track record leading architecture and end-to-end delivery: from database tuning, GIS/routing algorithms, and real-time streaming to modern web UIs (React 19, TypeScript, Vite), Go/Bun microservices, Docker/Kubernetes infrastructure, CI/CD automation, and AI-accelerated workflows.

Technical Expertise

- **Languages & Core:** Go, TypeScript / JavaScript, Rust, Python, C#, C++, SQL, Bash
 - **Backend & Runtimes:** Go Backend, Node.js, Bun, .NET / C#, Hono, Express, Fastify, GraphQL, tRPC, oRPC, REST APIs, WebSockets, WebRTC
 - **Frontend & UI:** React 19, Next.js, Vite, SolidJS, Blazor, Tailwind CSS, Ant Design, TanStack Query, Wouter, HTML5/CSS3
 - **Databases & Data Engineering:** PostgreSQL, SQLite, MongoDB, Redis, Supabase, Prisma, Drizzle ORM, PostGIS, GIS / OSRM routing matrices, spatial indexing
 - **Simulation & Optimization:** Discrete-Event Simulation (SimPy), Constraint Satisfaction Problems (CSP), Combinatorial Scheduling & Routing, Zero-Allocation Data Structures
 - **DevOps, Cloud & Infrastructure:** Docker containerization, Kubernetes (K9s, Helm), GitHub Actions CI/CD, Terraform, Linux Administration, Nginx, Traefik, Nomad, Cloudflare
 - **AI & Developer Tooling:** Agentic AI Tooling (Antigravity, Claude Code, Cursor, Copilot), Biome, uv, Hyperfine, Lazygit, Tmux
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Professional Experience

Lead Software Engineer & Systems Architect

Celerum Ltd & PlanSea Solutions — Aberdeen, UK

February 2017 – Present

- **Cloud-Native Platform Architecture:** Architected and engineered **PlanSea v2** (cloud-native marine logistics and offshore supply vessel planning platform) unifying legacy monorepos into a modern Bun, Hono, React 19, and TypeScript web platform with Python/SimPy simulation and C# optimization engines as specialized background workers.
- **Optimization & Simulation Engines:** Developed discrete-event simulation models and constraint-solving scheduling engines for offshore decommissioning, vessel sharing, and complex cargo logistics across North Sea operations.

- **High-Performance Microservices & GIS:** Implemented zero-allocation Go microservices and GIS routing pipelines (OSRM approximation and spatial distance matrices) processing large-scale geospatial and AIS (Automatic Identification System) vessel telemetry data.
- **Database & Query Optimization:** Architected multi-tenant data tiers across PostgreSQL, SQLite, MongoDB, and Redis; designed optimized schema migrations, spatial indexes, and caching strategies delivering sub-millisecond query latencies.
- **Full-Stack Web Applications:** Built responsive, reactive enterprise web portals, dashboards, and scheduling tools utilizing React, Vite, Ant Design, Tailwind CSS, and WebSockets for real-time fleet tracking.
- **DevOps & CI/CD Infrastructure:** Designed containerized deployment pipelines using Docker, Kubernetes, and GitHub Actions; established automated linting, testcontainers, and fast-feedback builds reducing release deployment cycles.
- **Technical Mentorship & Standards:** Led engineering best practices, code reviews, architectural documentation, and supervised university R&D projects and junior engineers.

Software Engineer (KTP Associate)

Robert Gordon University & ARR Craib – Aberdeen, UK

September 2014 – February 2017

- **Fleet Management System:** Designed, developed, and deployed an enterprise-wide real-time fleet logistics and dispatch management system for road haulage operations.
- **Distributed Services & Scaling:** Engineered load-balanced microservices handling high-concurrency vehicle telemetry, automated job scheduling, and driver dispatching using Meteor.js, Node.js, MongoDB, and Java.
- **Leadership & Stakeholder Alignment:** Led user adoption, operator training, and workflow digitalization across depot networks; completed professional management and leadership training under the UK Knowledge Transfer Partnership (KTP).

Education

- **Bachelor of Science in Computer Science (Software Development for Mobile Devices)**

Université Blaise Pascal, Clermont-Ferrand, France (2013 – 2014)

- **Bachelor of Science in Computer Science**

IUT Clermont-Ferrand, France (2011 – 2013)

Peer-Reviewed Publication

- **An Analysis of Indirect Optimisation Strategies for Scheduling**

Charles Neau, Olivier Regnier-Coudert, and John McCall.

IEEE World Congress on Computational Intelligence (IEEE WCCI 2018).